UP-A-CREEK ROBOTICS · ENGINEERING PORTFOLIO

Season: PowerPlay (2022-23) | Location: Longmont, Colorado, USA
Organization / School: St. Vrain Valley School District

MAJOR AWARDS & JUDGING RECOGNITION

· World Championship Finalist Alliance · Houston Worlds
· Innovate Award Winner · Sponsored by Raytheon Technologies
· Think Award Finalist · Colorado Regional Championship

EXECUTIVE SUMMARY

FTC #11260 Up-A-Creek Robotics pioneered the continuous 360-degree turret in FIRST Tech Challenge, setting world-record scoring cycles through mechanical innovation and seamless driver integration.

CORE SUBSYSTEMS & MECHANISMS

· 360-Degree Continuous Planetary Turret
· Dual Carbon Fiber Elevator with Spring Assist
· Compliant Molded Silicone Cone Centering Gripper
· Ultra-Compact Low-CG Odometry Pods

MECHANICAL & CAD ENGINEERING HIGHLIGHTS

· Continuous 360° Turret: Custom ball-bearing race with slip-ring wire pass-through enabling infinite turret rotation while moving.
· Carbon Fiber Box-Tube Lift: Constant-force spring counterbalancing reducing motor draw during rapid 34-inch vertical extensions.
· Compliant Cone Centering Gripper: Molded silicone gripper pads accommodating severe cone misalignments during high-speed pickup.

AUTONOMOUS NAVIGATION & SOFTWARE

· Field-Centric Turret Slaving: Turret autonomously targets scoring junctions regardless of robot chassis orientation.
· On-The-Fly Dynamic Pathing: Adaptive trajectory recalculation avoiding defense robots while maintaining cycle tempo.
· Dual-Camera Computer Vision: Color thresholding and contour analysis detecting junction poles with 99.2% accuracy.

COMMUNITY OUTREACH & STEM ADVOCACY

CONNECTING & MOTIVATING INITIATIVES

· Mobile STEM Discovery Trailer: Traveled to rural communities across Colorado, engaging 3,000+ students in hands-on robotics activities.

· Aerospace Industry Mentorship Pipeline: Partnered with local aerospace manufacturers to provide student internships.

· Colorado FTC Scrimmages: Hosted preseason events and safety workshops for 24 regional robotics teams.